Fırat Atalay

Backend Software Engineer

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Professional Summary

Backend-focused Computer Engineering graduate with hands-on experience contributing to real-world software products through long-term and summer internships, part-time work, and team-based projects. Built **REST APIs**, authentication and authorization flows, multi-tenant applications, and database-backed systems using **ASP.NET Core**, **Entity Framework Core**, **Spring Boot**, **PostgreSQL**, and **Docker**. Experienced in collaborative development workflows involving task management, branching, pull requests, and code reviews, with additional full-stack experience in **React**, **TypeScript**, and **Vue.js**. Focused on building a long-term career in backend software engineering and contributing to reliable, maintainable, and well-structured software systems.

Technical Skills

Backend: C#, ASP.NET Core, Entity Framework Core (EF Core), RESTful APIs, Python, FastAPI, Java, Spring Boot

Databases: PostgreSQL, Microsoft SQL Server, MongoDB

Frontend: React, TypeScript, JavaScript, Vue.js, Next.js, Tailwind CSS, Material UI

Architecture & Practices: RESTful API Design, Layered Architecture, CQRS, Authentication & Authorization, Role-Based Access Control (RBAC), Multi-Tenancy

Messaging & Development Tools: RabbitMQ, Git, GitHub, Docker, Docker Compose, Flyway, Postman, Swagger/OpenAPI

Testing: xUnit, Jest, Vitest, Cypress

Professional Experience

Elasoft Trabzon, Turkey

Software Development Intern – Long-Term & Summer Placements

- Completed a 60-working-day summer internship and later returned for a full-semester industry placement, contributing to real-world software products across both periods.
- Developed backend and full-stack features for a university **Community Management System** and an internal **Human Resources Management System** using **C#**, **ASP.NET Core**, **Entity Framework Core**, **Microsoft SQL Server**, and **Vue.js**.
- Implemented and improved personnel, leave-request, profile and avatar, asset-assignment, role-based access, validation, and document-export workflows across backend APIs and frontend modules.
- Delivered assigned tasks through Git branches and pull requests, documented implementation changes, and incorporated feedback from senior software engineers through code reviews.

Medyanes 360

Volunteer Software Developer

- Selected after completing a full-stack technical case and presenting the solution during a live technical interview.
- Contributed to web applications across multiple business domains, including a language-learning platform, by implementing assigned frontend and backend features.
- Worked through task, issue, commit, and pull-request workflows while collaborating with the development team and incorporating project feedback.

Recep Tayyip Erdoğan University Rize, Turkey

Part-Time Software Developer

- Contributed to internal university information systems and administrative software by developing full-stack modules that helped digitize and streamline operational workflows.

- Collaborated directly with non-technical university staff to identify operational problems, translate requirements into software features, and refine solutions based on stakeholder feedback.

Selected Projects

Constructor ERP — Multi-Tenant Construction Site Management SaaS GitHub Repository

Live Demo — sign in as any of five roles without a password

Java 25, Spring Boot 4, PostgreSQL 18, Flyway, Vue 3, TypeScript, Docker

- Co-developed and deployed a multi-tenant SaaS product that lets contractors run their construction sites from one place instead of WhatsApp groups, covering site chat, attendance and timesheets, material shipments, subcontractor progress, subscriptions, and platform administration, with separate phone and desktop interfaces.
- Isolated tenant data with PostgreSQL row-level security (RLS) on top of service-level filtering, with a startup check that refuses to run if any table is unprotected; implemented server-side sessions, Sec-Fetch-Site-based CSRF protection, Flyway migrations, and a one-command Docker setup.
- Enforced file and function size limits and layer boundaries at build time with ESLint and Checkstyle, and built a browser-only live demo without touching screen code to work around free-hosting limits.

FraudCell — Event-Driven Fraud Operations Platform GitHub Repository

Turkcell Code Night 2026 All-Star Final

ASP.NET Core 10, YARP, FastAPI, scikit-learn, RabbitMQ, PostgreSQL, React, Docker

- Co-developed a containerized fraud operations platform comprising Identity, Transaction, AI, and Gamification services behind a YARP API Gateway, with a separate PostgreSQL database for each service and RabbitMQ-based asynchronous communication.
- Implemented event-driven transaction assessment using transactional outbox/inbox patterns, role-based authorization, real-time notifications, and fallback handling that kept transaction intake operational when the AI service was unavailable.

ZekAI — AI-Powered E-Commerce Decision Platform GitHub Repository

Live Demo — HackAthon'26; sign in as a buyer or a seller without a password

Next.js 16, React 19, ASP.NET Core 10, EF Core, PostgreSQL 16, FastAPI, Google Gemini, Server-Sent Events, Docker

- Captained a two-person team at HackAthon'26 (BTK Akademi × Google × Girişimcilik Vakfı) that built, in six days, an AI decision platform for e-commerce buyers and sellers plus the marketplace it integrates with: two Next.js apps, two ASP.NET Core APIs, a FastAPI service on Google Gemini, and PostgreSQL.
- Kept purchase verdicts deterministic by letting Gemini produce only scores and evidence while a rule engine decides; cut the buyer pipeline from seven agent calls to one structured call, cached results by input hash and product fingerprint, and streamed progress to the browser over SSE.
- Integrated with the marketplace through its API using seller-issued keys accepted alongside JWTs in one policy scheme, and later moved both frontends to a browser-only live demo behind the same API contract.

EduConnect — AI-Assisted Campus Platform GitHub Repository

Live Demo — TÜBİTAK 2209-A supported; sign in as any of four roles without a password

ASP.NET Core 10, EF Core 10, PostgreSQL 16, SignalR, FastAPI, BERTurk, Gemini, React 19, Docker

- Captained a two-person team that built a TÜBİTAK 2209-A supported graduation project: a campus platform with a feed, communities, events, a student marketplace, real-time messaging, and EduAI, an assistant answering from a 412-entry knowledge base built from official sources.
- Combined a fine-tuned BERTurk classifier, knowledge base retrieval, and Gemini behind confidence thresholds and fallbacks, so answers cite their source and keep coming when Gemini or the NLP service is down; resolved refresh-token and like races in the database.
- Moved the AI models to a separate service to fit a 512 MB free tier, migrated from SQL Server to PostgreSQL, and replaced the expired live site with a browser-only demo behind the same API contract.

Turkcell SkillHub — Gamified Digital Skills Learning Platform GitHub Repository

Live Demo — Code Night Trabzon 1st Place & All-Star Finalist

Java 25, Spring Boot 4, PostgreSQL 17, Flyway, React 19, TypeScript, Docker, Claude API

- Captained a two-person team to first place at Turkcell Code Night Trabzon, qualifying for the İstanbul All-Star Final, then rewrote the platform from scratch with a layered Spring Boot API, PostgreSQL, and a React/TypeScript interface, shipping a browser-only live demo and a one-command Docker setup.
- Modeled XP as an append-only ledger with rankings derived through SQL window functions, and graded quizzes on the server in a single transaction with row-level locking so that repeated or double submissions cannot farm XP.
- Integrated Claude for personalized recommendations, validating every suggestion against the learner's real progress with a deterministic rule-based fallback, and seeded realistic demo data by replaying learner histories through the real quiz service.

Smart Campus — Full-Stack Campus Ecosystem Management Platform GitHub Repository

Node.js, Express.js, PostgreSQL, Sequelize, React, TypeScript, JWT, Docker

- Co-developed and deployed a role-based campus ecosystem platform supporting authentication, GPS-based attendance, meal and wallet operations, event registration, classroom reservations, analytics, and notifications.
- Contributed to both the Node.js/Express backend and React/TypeScript frontend by implementing authorization and analytics endpoints, integrating module APIs, building authentication and profile flows, adding automated tests, and delivering interactive map functionality.

Education

Recep Tayyip Erdoğan University Rize, Turkey

B.Sc. in Computer Engineering *Sep 2021 – Jun 2026*

Awards & Technical Writing

Turkcell Geleceği Yazanlar Code Night

Trabzon 1st Place & All-Star Finalist | Team Captain

TÜBİTAK 2209-A University Students Research Projects Support Program

Funded Project Recipient | EduConnect

Huawei Student Developers

Volunteer Technical Writer

- Authored English-language technical articles on frontend and backend development for Huawei Developer's global and Türkiye publications, personal Medium, and Stackademic, with multiple articles exceeding 4K views and one surpassing 5K views.